MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

THIRD SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG3VACCFS200 – E-WASTE MANAGEMENT AND RECYCLING

Time:1.30 MaximumMarks:50

Part A

Short Answer Type Questions
Answer any **10** Questions
Each question carries **1** mark

1. Identify one valuable material found in E-waste. [Understand][1]
2. Summarize major environmental risk of landfilling E-waste. Understand][2]
3. Select advanced method is used to reduce the volume of E-waste physically. [Understand][2]
4. Identify the primary method used for recovering pure metals using bacteria. [Understand][4]
5. Predict what type of plants are typically used for phytoextraction. [Analyze][4]
6. Outline the name one trace metal that can be removed using phytoextraction. [Analyze][2]
7. Examine which enzymes help degrade pesticides in phytodegradation. [Analyze][3]
8. Illustrate the name a long-term benefit of using phytoremediation in urban E-waste zones. [Analyze][1]
9. Write one type of lighting equipment that becomes E-waste. [Apply][1]
10. Write which toxic metal is found in chip resistors and plastics. [Apply][1]
11. Select environmental advantage does adsorption offer in metal recovery. [Evaluate][1]
12. Evaluate oxidation in terms of electron transfer. [Evaluate][3]

Part B

Short Essay Type Questions
Answer any **4** Questions
Each question carries **5** marks

13. Explain the types of contaminants present in E-waste. [Understand][1]
14. Describe the process of extraction of nanometals from E-waste. [Understand][4]
15. Examine the limitations of phytoremediation as a method for treating E-waste contaminated sites. [Analyze][1]

16. Illustrate how phytoremediation is used for treating E-waste contaminated sites. [Analyze][3]
17. Write the processes of electrowinning and electrorefining in the recovery of metals from E-waste. [Apply][3]
18. Evaluate electrokinetic remediation, how it helps in the removal of organic contaminants from E-waste-contaminated soils. [Evaluate][4]

Part C

Essay Type Questions

Answer any **2** Questions

Each question carries **10** marks

19. Discuss the recovery of pure metals, formation of metal oxides, and the production of metal nanocomposites, along with relevant methods and applications. [Understand][4]

20. Illustrate the hydrometallurgical recovery methods used in the extraction of metals from E-waste. [Apply][3]

21. Evaluate the advantages, disadvantages, and challenges of using electrokinetic remediation for contaminated site treatment. [Evaluate][4]